Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: Lab02

Subject: **Operating System**

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1. For each of the following paths, identify whether it is absolute or relative:

Files path	Type
/home/user/Download	Absolute
../Report	Relative
/var	Absolute
Docs	Relative
/	Absolute

2. We have the tree directory structure:

```
sokmeak@Labtop-SokmeakSR:~/OS2024$ tree
.
├── etc
│   ├── networks
│   │   └── interfaces
│   ├── systemd
│   │   ├── resouled.conf
│   │   ├── system
│   │   ├── system.conf
│   │   ├── user
│   │   └── user.conf
│   └── udev
│       ├── rules.d
│       └── udev.conf
└── myhome
    ├── file01.txt
    ├── lost+found
    └── user
        └── Documents

12 directories, 6 files
```

- When the command ls -a are used then it will list all files in the udev directory including hidden files as well which is start with “.”

```
sokmeak@Labtop-SokmeakSR:~/OS2024$ cd /etc/udev
sokmeak@Labtop-SokmeakSR:/etc/udev$ ls -a
.  ..  hwdb.d  iocost.conf  rules.d  udev.conf
sokmeak@Labtop-SokmeakSR:/etc/udev$
```

3. Enter the shortest possible command for each of the following:
   a. Your current location is root (/). Enter the command to navigate to lost+found within the home directory.

```
sokmeak@Labtop-SokmeakSR:~/OS2024$ cd myhome/lost+found/
sokmeak@Labtop-SokmeakSR:~/OS2024/myhome/lost+found$
```

   b. Your current location is root (/). Enter the command to navigate to the directory named /etc/network/.

```
sokmeak@Labtop-SokmeakSR:~/OS2024$ cd etc/networks
sokmeak@Labtop-SokmeakSR:~/OS2024/etc/networks$
```

   c. Your current location is /home/user/Documents/. Navigate to the directory named /etc/.

```
sokmeak@Labtop-SokmeakSR:~/OS2024/myhome/user/Documents$ cd ../../../etc
sokmeak@Labtop-SokmeakSR:~/OS2024/etc$
```

   d. Your current location is /etc/systemd/system/. Navigate to the directory named /home/user/.

```
sokmeak@Labtop-SokmeakSR:~/OS2024/etc/systemd/system$ cd ../../../myhome/user
sokmeak@Labtop-SokmeakSR:~/OS2024/myhome/user$
```

4. Consider now we are at Home directory (user directory is consider as home directory). Using linux commands you have learned to:
   a. Display the current directory

```
sokmeak@Labtop-SokmeakSR:~$ pwd
/home/sokmeak
```

   b. Change directory to Desktop

```
sokmeak@Labtop-SokmeakSR:~$ cd Desktop/
sokmeak@Labtop-SokmeakSR:~/Desktop$ pwd
/home/sokmeak/Desktop
sokmeak@Labtop-SokmeakSR:~/Desktop$
```

   c. Create a folder named TPOS

```
sokmeak@Labtop-SokmeakSR:~/Desktop$ mkdir TPOS
sokmeak@Labtop-SokmeakSR:~/Desktop$ ls
TPOS
```

   d. Change directory to Home directory

```
sokmeak@Labtop-SokmeakSR:~/Desktop$ cd
sokmeak@Labtop-SokmeakSR:~$ pwd
/home/sokmeak
sokmeak@Labtop-SokmeakSR:~$
```

   e. Change directory to TPOS using Absolute Path

```
sokmeak@Labtop-SokmeakSR:~$ cd /home/sokmeak/Desktop/TPOS
sokmeak@Labtop-SokmeakSR:~/Desktop/TPOS$
```

   f. Change directory to a folder Document in the username folder using Relative Path

```
sokmeak@Labtop-SokmeakSR:~/Desktop/TPOS$ cd ../../Document/
sokmeak@Labtop-SokmeakSR:~/Document$
```

5. Using linux commands you have learned to:
   a. Create folder in a Document folder and name it “TP01”

```
sokmeak@Labtop-SokmeakSR:~/Document$ mkdir TP01
```

b. Create text file name “file.txt”

```
sokmeak@Labtop-SokmeakSR:~/Document/TP01$ touch file.txt
```

c. Write “I am YOUR NAME” in file.txt.

```
sokmeak@Labtop-SokmeakSR:~/Document/TP01$ nano file.txt
sokmeak@Labtop-SokmeakSR:~/Document/TP01$ cat file.txt
I am SAREN Sokmeak.
```

d. Change directory to Desktop using Relative path.

```
sokmeak@Labtop-SokmeakSR:~/Document/TP01$ cd ..
sokmeak@Labtop-SokmeakSR:~/Document$
```

e. Read file “file.txt” from Desktop using Relative path

```
sokmeak@Labtop-SokmeakSR:~/Document$ cat TP01/file.txt
I am SAREN Sokmeak.
sokmeak@Labtop-SokmeakSR:~/Document$
```